

The Meal Loop

Closing the feedback loop in the municipal school meal service

KTH Royal Institute of Technology • ME2321 Engineering and Global Challenges • CONOPS Project
Group XX: Name 1, Name 2, Name 3, Name 4 • 2 October 2026

CONOPS steps and where they are covered. 1 system of interest: §1 • 2 current system, boundary and scale: §2 • 3 need: §3 • 4 nature of the changes: §4 • 5 alternatives and drawbacks: §5 • 6 proposed system, scope and objectives: §6 • 7 operational scenarios: §7 • 8 operational impacts: §8 • 9 organisational impacts: §9 • 10 SDG assessment: §10 • 11 trade-offs: §11

1 System of Interest and Its Wider Context

The system of interest (SOI) is the **municipal school meal service**: the planning, procurement, preparation, serving and consumption of school meals, and the handling of the food that remains after each meal. Its purpose is to provide pupils with sufficient meals in a way that is operationally reliable and resource-efficient. Food waste is an outcome of how the parts of the service interact.

The meal service is part of the larger municipal education system and depends on several external systems: food suppliers, pupil attendance and scheduling, and waste management. It is socio-technical because its performance depends on the interaction between pupils, kitchen staff, school and municipal organisations, planning and procurement routines, information systems, and physical infrastructure such as kitchens and dining facilities. No single technical component decides the result.

We chose this system because it is a large public service with a visible sustainability problem. Roughly one third of the food produced for human consumption is lost or wasted somewhere between farm and plate [2], and reducing food waste is the explicit aim of SDG target 12.3. Municipalities serve free meals every school day, and a large share is thrown away as plate waste, unserved food from the buffet, or kitchen scraps. The cost is both environmental and financial, since the municipality pays for ingredients, staff, facilities and waste collection, and all of it is lost when food goes in the bin. The system is also complex enough to be worth analysing: procurement rules, kitchen routines, school schedules and pupil behaviour all meet in the same daily operation.

2 The Current (Predecessor) System

2.1 Function

The primary function is to provide pupils with a meal during the school day, delivered in three phases. **Before the meal**, the system plans menus, estimates demand, procures food and allocates resources, which decides what is available and in what quantity. **During the meal**, kitchen staff turn those resources into meals, and the serving arrangement and the pupils' own choices decide how much is actually eaten. **After the meal**, the system handles what was not eaten, both prepared surplus and plate waste; this phase could also produce information about consumption, which would show how well planning and production worked.

2.2 Main Components and Their Relations

Table 1: Main components of the current school meal system.

Category	Main components
Actors	Pupils, kitchen staff, school management, municipal meal planners, procurement staff
Organisations	Schools, school kitchens, municipal meal-service functions, municipal administration
Technical components	Menu-planning and ordering systems, scheduling and attendance systems, waste recording where it exists
Infrastructure	Kitchens, storage, serving areas, dining spaces, preparation and serving equipment
Information flows	Pupil numbers, schedules, menus, expected demand, ordered quantities, consumption and waste data

The components are interdependent. Planning and procurement decide how much food is available, kitchen staff decide how it is prepared and presented, and attendance, food choices and portioning then decide how much is eaten and how much remains. The system therefore consists of two flows that do not behave the same way: the physical flow of food runs forward and is relatively direct, while information about actual consumption and waste does not flow back with the same consistency. This separation between the material flow and the information flow is what the rest of the analysis builds on.

2.3 System Boundary and Its Motivation

Table 2: System boundary: what is inside the analysis and what is not.

Dimension	Inside	Outside, and why
Organisation and place	The municipal compulsory-school meal service, with one school as an illustrative unit	The national school meal system as a whole, which is too broad for this project
Actors	Pupils, kitchen staff, school management, meal planners, procurement staff	Food suppliers and waste contractors, which interact with the system but are treated as external
Processes and information	Menu and demand planning, procurement, preparation, serving, consumption, handling and measurement of surplus and plate waste, and the data connecting them	Agriculture and food manufacturing upstream, waste treatment downstream, and pupils' food consumption outside school
Time	The daily meal cycle, the menu cycle and the school term	Long-term national policy cycles

Motivation. The boundary is drawn around the activities and information flows that directly influence how much food is planned, made available, consumed and finally wasted. The municipal level is broad enough to include the interaction between schools, kitchens and municipal functions, but limited enough to analyse in this project. A tighter boundary, a single kitchen, would make the problem look like a local cooking error although the causes are spread across planning, scheduling and serving. A wider boundary, the whole food system, would make it impossible to say who should act. Since the core processes recur across compulsory schools, one school works well as an illustrative unit.

2.4 Scale, Scope and Temporality

The system works on two organisational levels: meals are prepared, served and discarded in individual schools, while menu planning, procurement, budgeting and routines are coordinated at municipal level. It also works on three time horizons, and this is where much of the problem sits. Food is produced and discarded **daily**; menus, demand estimates, purchasing and staffing are decided over

weeks or months; information about waste accumulates over **terms**. Decisions before the meal rest on expectations about future demand, while actual consumption is only known during and after it, and this delay limits how well the system can adapt or use what it learned yesterday.

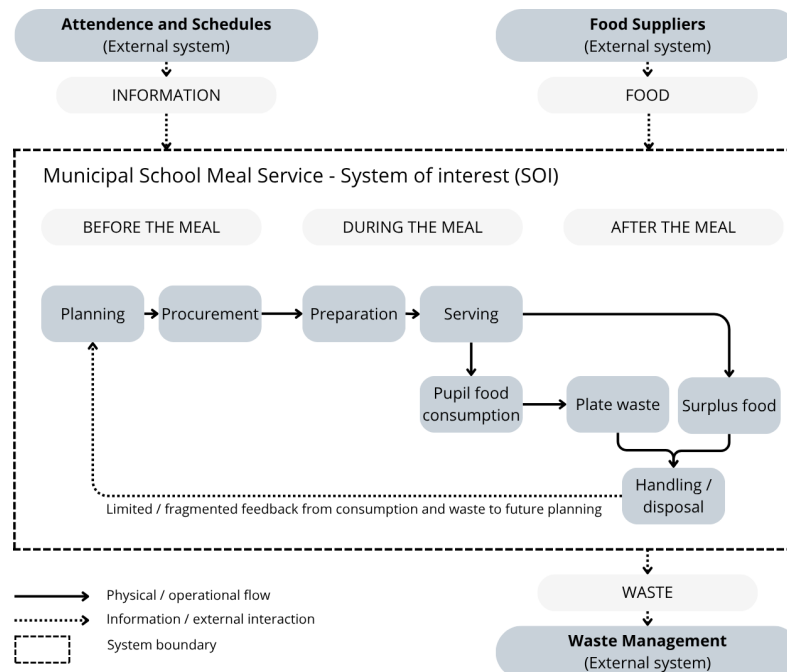


Figure 1: The current system: a direct forward flow of food, and only limited and fragmented feedback from consumption and waste back to planning.

3 Need and Motivation for Change

3.1 Identified Need

The need is simple to state: the school meal service should waste less food, and it should do this without giving pupils less to eat.

Today the service works as a largely linear open-loop “push” system. What is produced is decided in advance and pushed out to the dining hall, and what is eaten has no effect on what is produced next. Procurement, kitchens and schools each work well on their own, but the result for the service as a whole is poor, which is what the *suboptimisation principle* describes [3]. The over-production that follows is the food waste. What the service needs is a feedback loop, so that the amount actually eaten reaches back to the people who decide how much is cooked and how much is ordered.

Problem statement: School kitchens cook more food than the pupils eat. Meals are planned and ordered from the number of registered pupils instead of the number who actually eat, and nobody measures what is left over. The same over-production is therefore repeated day after day, which costs the municipality money and causes unnecessary environmental impact from food that nobody eats.

3.2 Why the Waste Happens

The waste is not one mistake in one place. It arises in all three phases of the meal, for a different reason in each.

- **Before the meal.** Menus are set weeks in advance and food is ordered from the number of pupils registered at the school, not from the number who will actually eat. Absence, school trips and unpopular dishes are not taken into account.
- **During the meal.** Nobody measures how much is cooked, how much is put out and how much is left. Serving pans are filled generously so the line never looks empty, and food that has been on

the buffet cannot be put back and used later.

- **After the meal.** What is left is thrown away, and the information about it stays in the kitchen. Next week's menu and next week's order are made from the same numbers as before.

Behind all three lies one common cause: nobody is responsible for the result as a whole. The school owns the timetable, the planner owns the menu, and the kitchen is the one blamed for the waste. Each of them does their own job well, but the waste belongs to everyone and therefore to nobody.

3.3 Why Change Now

Food waste is a national priority in Sweden and is connected to SDG 12 on responsible consumption and production, so a municipality that reduces it can show progress on goals it has already committed to. The waste is also expensive. The municipality pays for ingredients, working hours and waste collection for food that gives nutrition to nobody, and that money comes from the same budget as the meals themselves.

3.4 Stakeholders and Their Needs

A socio-technical system is defined partly by its stakeholders and their different, sometimes conflicting, worldviews [1].

Table 3: Key stakeholders, their interests and their concerns.

Stakeholder	Main interests / needs	Possible concerns
Kitchen staff	Predictable workflows; less overproduction and less of the burden of throwing away edible food	Running out of food; stress from late changes to production targets
Pupils	Enough nutritious and appealing food; a positive dining environment	Empty trays at the buffet; feeling restricted or blamed for waste
School administration	Smooth logistics; staying within budget; meeting nutritional requirements	Extra administrative work; friction with parents if the service looks inadequate
Municipality and society	Climate targets; efficient use of tax-funded services	High upfront cost for new IT; risk of breaking food-safety rules

3.5 Aim and Project Question

The aim is to develop and evaluate a concept of operations for a municipal school meal service in which what is measured after the meal informs how much is cooked and served during it and what is procured before it, and to assess the consequences of such a change for the stakeholders involved.

How can a municipal school meal service connect planning, production and waste handling into one feedback loop so that less food is wasted, and what are the operational, organisational and sustainability consequences of such a change?

4 Nature of the Proposed Changes

4.1 The New Features

We propose four changes to the way the meal service is run. Together we call them the Meal Loop, and the system itself is described in Section 6.

1. **A morning count of eaters.** The school reports absences, trips and other known changes to the kitchen in the morning, and the kitchen cooks for that number instead of for the class list.
2. **Two-step cooking.** Only a first part of the food is cooked and put out. The rest stays in the kitchen and is released only if the serving line needs it.
3. **Waste recorded per dish.** What is left is weighed and written down together with the dish and the weekday, instead of only as a weekly total.

4. **A shared waste target.** The school and the kitchen get the same target, and the meal manager in the municipality is responsible for following it up.

4.2 What Kind of Change Is This?

Is it new technology? Almost none. The information the system needs already exists, but in separate places: absences are reported every morning in every school, and most kitchens already have scales. What is missing is not a tool but a routine for passing the numbers on and using them. The only new element is a record of waste per dish, which can be a sheet on the wall or a field in the system the kitchen already uses. We chose this on purpose. An app or a sensor solution would cost more, would have to be procured, and would collect data about individual pupils, which we want to avoid.

Is it a new business model? No. The meal stays free for the pupils and funded by the municipality, and the suppliers and contracts are the same. What changes is how the money that is already being spent is used, since less of it goes to food that is thrown away.

One sub-system, or all stakeholders? It is not one sub-system. The change follows the whole chain, from planning and ordering through cooking and serving to what happens with the leftovers, and every stakeholder group in Table 3 is affected, but not equally:

- **kitchen staff** get the largest change in their daily work, since they cook in two steps and record what is left;
- **the school** has to send the morning count, and has to accept that the timetable and attendance are part of the meal service;
- **menu planners** have to use the waste records when they plan the next menu;
- **the municipality** has to set the target and follow it up;
- **pupils** notice the least. They still choose from a buffet, but the pans are smaller and are refilled.

4.3 Justification

Each change answers one of the reasons for the waste in Section 3. The morning count is there because the kitchen, otherwise, cooks for pupils who are not at school. Two-step cooking is there because food that has been on the buffet cannot be reused, so it is better to keep the rest in the kitchen where it still can be. Recording waste per dish is there because a weekly total does not tell the planner which dish failed. The shared target is there because the waste is nobody's responsibility today, and a routine that nobody owns will not survive a busy lunch.

The changes are also chosen so that they can actually be carried out. They use the people, rooms and equipment the service already has. They do not conflict with the food-safety rules, because what is held back is food that has never been served. And they can be tried in a few schools before a whole municipality changes.

5 Concept Development and Alternatives

5.1 Methods Used

The concept-development process followed an iterative approach, drawing on systems thinking and the divergence–convergence structure of the Double Diamond model. Mapping the current system and its stakeholders helped identify possible points of intervention beyond individual pupil behaviour. Brainstorming was used to generate alternative interventions without immediate evaluation. The resulting concepts were subsequently examined in relation to the identified system needs, stakeholder interests, and operational constraints. Particular attention was given to food availability, resource efficiency, and implementation feasibility.

5.2 Generation of Alternatives

The initial exploration considered reducing food waste by influencing pupil behaviour. However, analysis of the predecessor system highlighted that waste arises at several stages, involving different actors and decisions. The scope of potential interventions was therefore expanded to include measurement, demand estimation, and operational coordination.

Three targeted approaches were identified, followed by a fourth concept combining selected mechanisms into a broader socio-technical intervention:

- **Concept A – Behavioural intervention.** Encourage pupils to take appropriate initial portions, with the possibility of refilling. This primarily addresses plate waste but has limited influence on upstream planning and overproduction.
- **Concept B – Improved waste measurement.** Introduce systematic, dish-specific measurements of prepared surplus, serving waste, and plate waste. This improves observability, but measurement alone does not ensure that the information influences subsequent decisions.
- **Concept C – Demand forecasting.** Use attendance information and historical consumption patterns to improve estimates of daily meal demand. This may reduce overproduction, but forecasting uncertainty remains, and predictions alone do not address serving routines or the handling of surplus.
- **Concept D – The Meal Loop.** Integrate updated morning demand counts, flexible two-step cooking, dish-level waste measurement, and shared operational responsibility. This concept combines improved information with changes to production and decision-making, but requires additional coordination and adjustments to existing workflows.

Concept D was thus developed as a combination of elements from concepts B and C.

5.3 Evaluation and Selection

The concepts were compared against the identified system needs by considering their potential to reduce waste, respond to changing demand, use information for future decisions, maintain meal availability, and be implemented within existing operations.

Table 4: Qualitative comparison of alternative concepts.

Criterion	A	B	C	D
Potential to address multiple waste sources	Limited	Indirect	Partial	Broad
Response to same-day demand variation	Limited	None	Before meal	Before/during
Use of waste data in future decisions	No	Possible	Optional	Integrated
Implementation effort	Low	Low-medium	Variable	Medium
Meal availability	Unchanged	Unchanged	Forecast-dependent	Maintained through additional batches

Concept D, the Meal Loop, was selected for further development because it combines improved demand information with the ability to adjust production and use measured outcomes in future planning.

Rather than treating the alternatives as mutually exclusive, the selected concept incorporates elements of Concepts B and C while introducing changes to responsibilities and kitchen routines. Concept A remains a possible complementary measure, particularly for addressing plate waste.

6 The Proposed System: The Meal Loop

6.1 Overview, Scope and Objectives

The proposed Meal Loop retains the same overall system boundary as the current system, (Table 2), but changes how information is used to coordinate planning, production, serving, and waste handling, and keeps the same organisations. There are four main changes:

- **updated daily demand information**, where attendance and scheduling information is used to revise the expected number of meals before production;
- **two-step production**, where an initial quantity is prepared and additional batches can be produced in response to observed demand during the meal;
- **systematic waste measurement**, where plate waste and other relevant waste outcomes are recorded and used as feedback for future planning;
- **handling of surplus food**, where surplus is assessed for safe internal reuse or recovery before the remaining waste leaves the system.

The objective is not to minimise food production as far as possible, but to better match production with actual demand while preserving the function of the system which is providing sufficient meals to pupils. Hence, the proposed system seeks to reduce overproduction and waste without removing the ability to maintain meal availability.

The concept also introduces a stronger connection between daily operation and longer-term planning. Information generated during and after the meal is used to improve future planning decisions. By distributing responsibility for waste reduction across relevant actors in the system, the concept establishes waste reduction as an organisational principle rather than it being treated as an isolated technical component.

6.2 Structure and Functions

The proposed system combines physical food flows with information and control flows operating over different time horizons. Before the meal, long-term planning and procurement establish the resources available for meal production. Attendance and scheduling information from the surrounding school system is then used to produce a *daily demand update*, which adjusts the expected demand for that particular day. This information influences the quantity prepared in the initial production stage.

During the meal, food is *served in batches*, instead of committing the full expected quantity at once. The kitchen still retains the ability to prepare additional batches. Actual demand is observed as the meal proceeds, and this information forms a short-term control loop. The demand is observed and communicated to prepare additional batches, which in turn replenishes the food when required.

After the meal, two different outcomes are distinguished. Food taken by pupils but not consumed becomes *plate waste*, which is measured before entering waste handling. Food that remains as unserved surplus is instead subject to a *reuse/recovery assessment*. Where food may remain within the school meal system if safe. Food that cannot be reused is transferred to the external waste-management system.

The proposed system also contains a long-term feedback loop. Information from demand observation and waste measurement is returned to long-term planning. This information can allow for better forecasting of future needs, menu planning and other related decisions.

The Meal Loop therefore performs two forms of adaptation. The *same-day operational loop* responds to variation during an individual meal through additional batch preparation, while the *long-term learning loop* uses accumulated demand and waste information to improve future planning. Together, these loops connect the before, during, and after phases of the meal service.

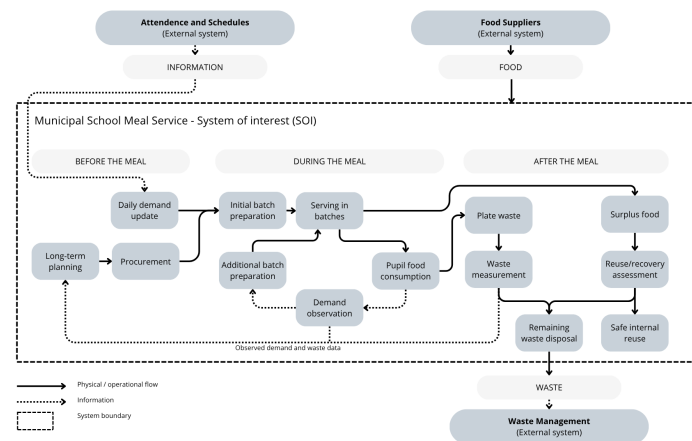


Figure 2: Representation of the proposed Meal Loop. The system combines updated daily demand information, adaptive batch-based production, separate handling of plate waste and surplus food, and feedback from observed demand and waste outcomes to support both same-day adjustment and longer-term planning.

Table 5: Comparison of the current and proposed school meal systems.

Dimension	Current system	Proposed system
Structure	Linear process with limited feedback between planning, production, consumption, and waste handling.	Connected process with short-term adjustment and long-term feedback.
Demand and production	Production is mainly based on demand estimates made before the meal.	Daily demand information informs initial production, while additional batches can be prepared in response to observed demand.
Information and feedback	Information about actual consumption and waste is fragmented or may reach planning only indirectly or with delay.	Observed demand and measured waste are systematically returned to future planning and production decisions.
Waste handling	Plate waste and surplus food are handled primarily after they have already arisen, with limited differentiation in subsequent use.	Plate waste is measured, while surplus food is assessed separately for safe reuse or recovery before residual waste leaves the system.
Roles and responsibility	Responsibilities are distributed across different parts of the meal service, with limited shared accountability for total system waste.	Waste reduction becomes a shared organisational responsibility.

7 Operational Scenarios

Not written yet. The three perspectives below come from the stakeholder table in Section 3. For each one, relate the system functions to a mode of operation (normal day, exam day, school trip, or a school that only reheats food), a class of user, and the external systems involved (attendance, suppliers, waste management). Write them as desired-to-be situations, roughly a third of a page each. If a scenario reveals an awkward consequence, keep it and carry it into Section 10.

7.1 Scenario 1 — Kitchen staff

Situation. *[A normal school day: the menu, the number of pupils on the list, what the morning note says.]* **Interaction.** *[How the kitchen uses the count and cooks in two steps.]* **System response.** *[When the second part is released, and what is recorded.]* **Outcome.** *[The result for the staff, including what is still difficult.]*

7.2 Scenario 2 — Pupils

Situation. [...] **Interaction.** [...] **System response.** [...] **Outcome.** [...]

7.3 Scenario 3 — Municipality and society

Situation. [...] **Interaction.** [...] **System response.** [...] **Outcome.** [...]

8 Operational Impacts

Partly seeded from the stakeholder concerns in Table 3; the empty cells still need writing. Keep the two time frames apart, since the step asks for impact both during transition and after implementation.

Table 6: Operational impacts by group and time frame.

Group	During transition	After implementation
Pupils	[-]	The same free meal and buffet, with smaller pans that are refilled
Kitchen staff	A new routine to learn; stress from changing production targets late	Less overproduction and less edible food thrown away; more coordination during service
School administration	Morning reporting must reach the kitchen reliably	One extra administrative step, but no new system to operate
Municipality	[-]	Ordering follows real demand; no large IT investment
Support and society	[-]	[-]

9 Organisational Impacts

The points below follow from the concept in Section 6 and still need to be developed, above all for the development and transition phase.

- **Responsibilities.** Waste reduction becomes a joint target for school and kitchen instead of a kitchen matter; the meal manager owns the loop and the planner has a standing duty to use the dish data each cycle.
- **Workflows.** The morning count and the two-step release become normal parts of the daily routine.
- **Coordination.** The school office and the kitchen need a reliable link on the same morning; the concept depends on it.
- **Competencies and decision-making.** *[What staff need to learn, and which decisions move closer to the operation.]*

Brief sustainability consequence assessment. *[Social, environmental and economic consequences at organisational level. One point worth making: the target must include food eaten and not only food thrown away, otherwise the system can report success while pupils simply eat less.]*

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10 Trade-Offs, Benefits and Limitations

The trade-offs below come from the stakeholder concerns in Table 3. Finish the table, then discuss the one or two hardest trade-offs in prose, since the step asks for analysis and not only a list.

Table 7: Main trade-offs associated with the Meal Loop.

Benefit / objective		Cost or disadvantage accepted
Cooking follows actual demand, so serving waste falls	↔	Higher risk that pupils meet empty trays late in the service
The kitchen holds food back as a buffer, so less becomes unusable	↔	More coordination and more stress during a short lunch service
Dish-level data gives planners facts instead of impressions	↔	Extra recording work and administrative overhead
Existing lists and scales are reused, so no large IT investment	↔	<i>[-]</i>

Risks. Operationally, a wrong morning count can empty a dish before the end of the service. Organisationally, waste figures can be used to blame the kitchen, or the silos can refuse a shared target. In regulatory terms, food-safety and hygiene rules are breached if held-back food is handled incorrectly. *[Add the economic risk that recording time costs more than the food saved, and the user-related risk that pupils feel restricted or blamed.]*

Benefits, limitations and disadvantages. *[To be written: what the concept achieves that the current system cannot and for whom; where it is weaker; and the limitations of the analysis itself, for example that no waste data were collected in a school, that stakeholder behaviour is assumed, and that this is a concept rather than a tested implementation.]*

11 Conclusion and Reflection

Summarise the need, the concept, its main expected benefits and its main trade-offs, then reflect on the methods: which were most useful, whether they changed your understanding of the problem (the shift from “careless pupils” to a wicked problem is a good example), and what you would do differently with more time.

References

- [1] Peter Checkland. *Systems Thinking, Systems Practice*. John Wiley & Sons, 1981.
- [2] Jenny Gustavsson, Christel Cederberg, Ulf Sonesson, Robert van Otterdijk, and Alexandre Meybeck. Global food losses and food waste: Extent, causes and prevention. Technical report, Food and Agriculture Organization of the United Nations, Rome, 2011.
- [3] Ludwig von Bertalanffy. *General System Theory: Foundations, Development, Applications*. George Braziller, 1968.